higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 4

1 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1: PLASTICS

1.1 1.1.1
- Tar
- Tortoise shell
- Animal horn
- Some kinds of tree sap (Any 2 × 1)

1.1.2
- Vulcanised rubber
- Resins
- Celluloid
- Plastic used to make containers (Any 2 × 1)

(2 × 2) (4)

1.2
- Financially plastics are much cheaper to manufacture
- Plastics are recyclable
- Easy to manufacture
- Lighter than other material
- Corrosion resistant
- Have a high strength to weight ratio (Any 5 × 1) (5)

1.3
- Roofing
- Canopies for bakkies
- Plant pots
- Storage tanks
- Pipe's and sewers
- Bath, jacuzzies and swimming pools
- Boats, sports cars, aircraft (Any 6 × 1) (6)

1.4 Solvent welding is done at room temperature.✓ A solvent is applied to the surface of both plastics that will be welded.✓ It dissolves the surface material and frees the polymeric chains in both plastics.✓ The two surfaces are then pressed together, which allows their polymer chains to entangle with one another.✓ When the solvent has evaporated, the dissolved plastic hardens and the weld is completed.✓ (5)

[20]

QUESTION 2: ELASTOMERS

2.1
- Elasticity
- Snap/rebound
- Resilience
- Modulus of elasticity (Any 3 × 1) (3)

2.2
- Seals and gaskets
- Waterproofing
- Straps
- Tubing
- Protective clothing (Any 2 × 1) (2)

2.3
- Use a grinder or wire brush to✓ clean the surface to be sealed of all dirt particles, rust and previously sealing agents.✓
- If a primer is specified by the manufacturer, the specified primer is applied to the cleaned surface.✓
- Using a caulking gun apply the sealant and force it into the joint in the masonry or concrete.✓
- Leave the sealant to dry and harden.✓

(5)
[10]

QUESTION 3: ADHESIVES

3.1 If the surface is dirty and dusty, the dirt will prevent the adhesive from wetting and streaking to the surface.✓ If the surface is not sound and well fitted, the adhesive will have a little to steak to,✓ as they will be a gaps that adhesive cannot fill.✓ (3)

3.2
- White glue
- Yellow glue (2)

3.3
- It is non-toxic
- It is easy to apply
- It can be used indoors and outdoors
- It has good chemical resistance
- It produces a clear finish
- It has a long shelf life
- Adheres well and are flexible
- It is inexpensive
- It is nonflammable (Any 5 × 1) (5)

3.4
- Flexible
- Water-resistant
- Solvent
- Resistant to oil and chemicals
- Strong (5)

[15]

QUESTION 4: PAINTS, WATER REPELENTS, COATINGS AND PRESERVATIVES

4.1
- Review the manufacturers safety data sheet (MSDS) and technical data for the product you will be using.
- Always wear a mask.
- You should wear overalls to cover your clothes and as much of your body as possible.
- Wear safety glasses to cover your eyes.
- Wear gloves.
- Work in a well-lit and ventilated area.
- Keep paint out of the reach of children. (Any 5 × 1) (5)

4.2
- Remove any grease with mineral spirits by scrubbing the surface with a brush.
- If the surface is pitted or corroded use emery paper to smooth out the surface.
- Wash the object with soap and water to remove dust, paint and emery residue and rinse thoroughly.
- Wipe the surface with a moist rag of white vinegar to assure that it is ready to receive the primer.
- Spray or brush with a suitable primer and allow it to dry. (5)

4.3
- NFP: refers to the manufacturer (treatment company)
- 457: refers to SANS code
- S: refers to the end user i.e. structural
- 10: refers to the year of manufacture (treatment), i.e. 2010
- H2: refers to the hazard class i.e. the end use application (5)

4.4
- Environmentally safe as no toxic chemical is used
- Controllable and simple
- Operating costs are low
- Very little dust is produced
- It degreases and cleans at the same time
- It reduces or eliminates warping (Any 5 × 1) (5)

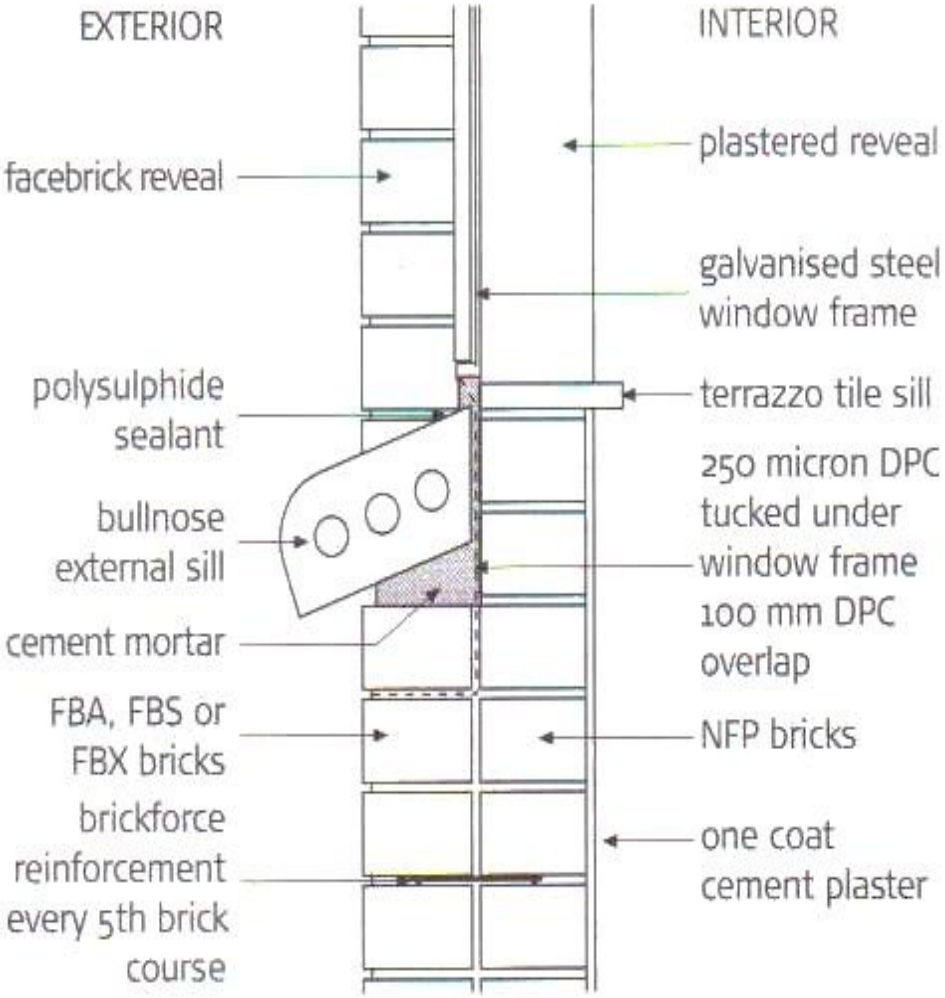
[20]

QUESTION 5: WATERPROOFING AND DAMP-PROOFING AND VAPOUR BARRIER INSTALLATION

5.1
- Remove any old waterproofing and clean the roof properly
- Put bond breakers over all joints in the concrete
- Apply primer to the dry concrete as this will allow the torch-on to stick
- Use a blow torch to melt the bitumen onto the roof
- Make sure that you allow enough overlap on all joints and that you smooth all overlaps with a trowel to ensure a smooth finish (5)

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5.2



Neatness = 1

Line-work = 2

Correctness = 2

Labels = 0,5 each (10)
[15]

QUESTION 6: BITUMEN, TAR AND ASPHALT

6.1
- A full length one piece overall of which the legs cover the boots.
- A face shield in cases where hot bitumen are worked with.
- Thick soled steel tip boots.
- A leather apron.
- Thick leather gloves.
- A SABS approved respirator
- 25-litre of clean water (5)

- A first aid kit specifically prepared for burns (Any 5 × 1)

6.2
- It pollutes the environment.
- It is highly volatile and may even explode.
- It is flammable and release toxic and flammable volatiles when burned.
- It can be dissolved in a wide range of solvents and is slightly soluble in water and can pollute the environment. (Any 2 × 1) (2)

6.3
- Penetration test
- Softening point test
- Viscosity test
- Ductility test
- Tests to determine the effect of heat and air. (Any 3 × 1) (3)

[10]

QUESTION 7: BITUMEN, TAR AND ASPHALT.

7.1
7.1.1 Self-tapping screw
7.1.2 Sheet metal screw
7.1.3 Socket screw
7.1.4 Set screw

(4 × 1) (4)

7.2
- To maintain quality control.
- To specify the type of bolt used on a specific part of the structure.
- To specify the tension required in certain sections of the structure.
- To indicate the number of bolts required to fix the structural joint.
- To reduce the possibility of wrong bolt usage.
- To provide a record of bolt used in the structure.
- To identify the material of the bolt.
- To provide torque or force indication specification. (Any 4 × 1) (4)

7.3
- Edge tear-out
- Bolt shear
- Bearing failure
- Net section failure (Any 2 × 1) (2)

[10]

TOTAL: 100

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